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THIRD SEMESTER B. TECH EXAMINATION, DECEMBER-2024

MATERIAL SCIENCE & ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	With a neat sketch define substitutional point defect.	L2	CO1	**02**
	b.	What are the different methods of measuring hardness of materials.	L2	CO2	**02**
	c.	Draw the microstructure for "Pearlite Structure of Steel".	L3	CO3	**02**
	d.	List any three objectives of hardening of materials.	L2	CO4	**02**
	e.	List down different types of composite materials based on the matrix materials.	L2	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Discuss the factors affecting diffusion.	L2	CO1	**05**
		OR			
	b.	Explain how the structure, properties and processing of materials are interconnected and discuss their implication on materials performance.	L2	CO1	**05**
3.	**a.**	Define fatigue loading? Define range of stress & stress amplitude.	L4	CO2	**05**
		OR			
	b.	Define creep and with a neat sketch explain the various stages of creep failure.	L2	CO2	**05**
4.	**a.**	What is homogeneous nucleation? Write its key characteristics?	L2	CO3	**05**
		OR			
	b.	List and explain two different types of corrosion in detail.	L2	CO3	**05**
5.	**a.**	Write short notes on the properties and uses of SiC and Al_2O_3.	L2	CO5	**05**
		OR			
	b.	Differentiate between ferrous and nonferrous materials.	L2	CO4	**05**
6.	**a.**	Explain the process of annealing and its effect on metal microstructure.	L2	CO4	**05**
		OR			
	b.	Differentiate between hardening and hardenability.	L4	CO3	**05**

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7.	**a.**	Name and explain two different types of line defects.	L2	CO1	**05**
		OR			
	b.	What is repeated and cyclic loading? How does it affect materials properties?	L4	CO2	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Define atomic packing factor. Calculate the atomic packing factor of BCC and FCC structure.	L2	CO1	**10**
		OR			
	b.	State and explain Fick's 1st law and 2nd law of diffusion. Also discuss the factors that affect diffusion.	L2	CO1	**10**
9.	**a.**	Explain the CUP & CONE fracture with the help of a neat sketch.	L2	CO2	**10**
		OR			
	b.	Draw and explain the various stages of stress strain curve for mild steel specimen.	L2	CO2	**10**
10.	**a.**	Explain the types of binary phase diagram and highlights the key features and differences.	L2	CO3	**10**
		OR			
	b.	Define the process of quenching. Mention its requirements in case of material processing? List down various quenching mediums.	L2	CO3	**10**
11.	**a.**	Describe the need for heat treatment of materials. Differentiate between hardness and hardenability.	L2 & L4	CO4	**10**
		OR			
	b.	Write short notes on: 1. Annealing 2. Normalizing 3. Nitriding 4. Boronizing	L2	CO4	**10**
12.	**a.**	Differentiate ceramic materials different from metals and Polymers? List four basic properties of ceramic materials.	L4	CO5	**10**
		OR			
	b.	Discuss the role of alloying elements in stainless steel and their impact on corrosion resistance and strength.	L4	CO4	**10**
13.	**a.**	Name the different types of Fibers and particle alignment in polymer matrix composites? Explain them with diagram.	L2	CO5	**10**
		OR			
	b.	Mention the importance of Ni-Ti alloy with respect to current industrial application.	L2	CO5	**10**